

Testing Theories of Interbranch Relations Using a Textual Measure of Congressional Delegation*

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Abstract

Congressional oversight of the federal bureaucracy remains key to understanding implementation of the law. Essential to this are theories of how and why Congress delegates powers to administrative agencies. Using an active learning convolutional neural network on bill text, we classify bill sections by their role in delegating to administrative agencies, applying an iteratively improving coding scheme that enhances existing supervised learning approaches. We systematically study the statutory scope of administrative agencies and develop a first-of-its-kind dataset to study how executive agency characteristics and changes in presidential administrations affect congressional decisions about delegation. First, we present descriptive data on the distribution of delegation in introduced bills across cabinet-level agencies. We then find evidence that variation in delegatory activity is driven by cross-agency differences in ideology and structural independence, even when conditioning on the party of the president. Our findings establish specific conditions for the operation of the ally principle, reflective of the executive branch's internal complexity.

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Delegation of powers from legislators to executive branch agents tasked with enacting policies remains at the heart of the challenges of modern governance. While delegation allows legislatures to benefit from policy expertise housed in the executive branch, it also may create principal-agent problems and allow for bureaucratic drift in policy-making and implementation. The potential for bureaucrats to subvert legislative policy preferences is not just a narrow issue concerning Congress; it also calls into question the extent of electoral accountability and responsiveness in a system of separated powers (Horn & Shepsle 1989). Given these stakes, it is important for political scientists to understand congressional decisions about delegation to the executive branch, as well as congressional capacity and willingness to oversee the exercise of delegated powers.

While the study of delegation in political science has been theoretically rich, large-scale empirical work has been hindered by measurement problems. Delegation is fundamentally a textual act: the legislature restricts or enables an agency in its actions through written law and legislative history. Consequently, even the most rudimentary measurement of delegation has to grapple with the written word, either directly or indirectly, to capture what is occurring. In any individual piece of legislation, this task is straightforward; identifying the agency and the change to their authority requires only a close reading of the bill. However, this approach is too time-intensive for large-scale applications and empirical studies.

Innovations in machine learning and natural language processing simplify scaling up these connections. In particular, these methods can predict qualitative labels from quantifiable features. After being trained on a set of hand-labeled examples, the learning algorithm is deployed to predict labels for a much larger data set, significantly reducing the up-front costs of setting up a human-run labeling project. However, most machine learning operations separate the hand-coding from the evaluation process and treat it merely as a single attempt at attaining accuracy. Complex concepts will be hard to translate into a simple coding scheme; thus, machine coding done this way can have additional problems.

In this paper, we argue that researchers can leverage their expertise through an inter-

active machine-learning framework that will essentially learn on the go. More commonly referred to as an “active learning” approach to classification, we argue that this framework best combines the portability and flexibility of machine learning with the expert evaluation usually used in more modestly defined approaches, such as case studies.

To demonstrate the utility of this framework, we tackle a canonical problem in political science: how congressional delegation decisions are affected by switches between unified and divided government. We create a new dataset on legislative delegation during the 110th and 111th Congresses—a period encompassing a change from divided government to unified government. Additionally, we use measures of agency ideology and agency independence to examine the workings of the ally principle more closely.

This paper proceeds as follows: first, we explain why delegation poses a unique measurement challenge. Second, we address this measurement problem by using a deep and active learning model, discussing what makes this different from other machine learning approaches. We then assess classification accuracy and discuss the performance of our active learning model, how it improves upon existing models, and what can be done with our newly labeled dataset. We conclude by using our measure of delegation to gain new leverage on the ally principle—the theoretical proposition that variation in delegation is explained by the degree of ideological disagreement between principal and agent.

Approaches to Measuring Delegation

As the table below illustrates, the empirical literature on delegation is characterized by a number of conceptual and methodological divides. While these studies generally share a broad definition of delegation as a grant of policy-relevant authority from the legislature to some other governmental body, they vary on how to operationalize this concept, where to look for it, and the appropriate unit of analysis at which to measure it. Some studies, such as Huber & Shipan (2002), evade the measurement of delegation altogether by selecting on

statutes that delegate and moving directly to the measurement of constraint and discretion.

Table 1: How Has Delegation Been Measured?

Citation	Sample	Method	Treatment of Delegation
Epstein & O’Halloran (1999)	257 Mayhew (1991) important laws	Close reading of law summaries	Legislative provisions that grant substantive policy discretion to some other governmental body
Huber & Shipan (2002)	67 state laws governing Medicare managed care (MMC) programs	Word count for legislation	Not explicitly measured, as the sample selects on laws that delegate
Farhang & Yaver (2016)	218 Mayhew (1991) important regulatory laws (updated through 2008)	Close reading of legislative text	Legislative provisions giving agencies the authority to perform regulatory functions
Bolton & Thrower (2019)	All executive agencies from FYs 1976-2015	Reading Appropriations subcommittee reports	New budget authority recommended for an agency in a given year
Anastasopoulos & Bertelli (2020)	59,423 EU laws (1958-2017)	Supervised machine learning framework with existing coded data	Same as Epstein & O’Halloran (1999)
Vannoni <i>et al.</i> (2021)	2,550 state laws (1900-2000)	Natural language parsing approach using syntactic parsing trees	Legal provisions with strictly defined syntax granting authority to some government actor
McCann & Shipan (2022)	443 Mayhew (1991) important laws (updated through 2016)	Use of ProQuest’s Regulatory Insight database, supplemented by close reading of legislative text	Evident in agency citations of law when promulgating rules/regulations

Of the studies that do explicitly measure delegation, most identify delegation at the legislative provision level and then aggregate up to the statute level (e.g., Epstein & O’Halloran 1999). However, one recent study (Bolton & Thrower 2019) measures delegation at the agency-year level rather than the statute level. Studies generally rely on text to identify and measure delegation, but there is variation in the types of text used—from full legislative text (Huber & Shipan 2002; Farhang & Yaver 2016; Anastasopoulos & Bertelli 2020; Vannoni *et al.* 2021) to legislative summaries (Epstein & O’Halloran 1999) to subcommittee reports

(Bolton & Thrower 2019).

Studies also vary in the extent to which they rely on human coders to identify and code delegation—and this methodological variation has important implications for the scalability of each approach. Furthermore, studies differ in the units they assign to measures of delegation, ranging from dollars to counts of delegating provisions. Finally, while most studies search for evidence of delegation in legislatively-produced texts, McCann & Shipan (2022) primarily rely on executive agency actions as indicators that authority has been delegated.

Our work here is the only application of automated methods to code delegation in U.S. legislation at the federal level. We also expand our focus beyond bills that become law, measuring delegation in each version of every introduced bill. Our measurement strategy is tied directly to legislative text, and we identify delegation and the recipients of delegated authority at the bill section level before aggregating up to the bill level.

Delegation, Constraints, and Discretion

With the exception of McCann & Shipan (2022), all of the work discussed above is interested in measuring delegation primarily as an intermediary step towards the measurement of another concept—discretion. Before describing our measurement strategy and methodological approach in detail, we differentiate between the interrelated concepts of delegation and discretion, and justify our decision to focus solely on delegation. Delegation is the decision made by a principal (in our case Congress) to assign a task to an agent (in our case, an executive branch actor). Discretion, on the other hand, refers to the amount of leeway an agent is given to carry out the task that has been delegated to them.

In some form or another, discretion is usually operationalized as the “amount” of policy authority that is left to an agency after taking into account any constraints the legislature includes in its delegation. These constraints typically come in the form of administrative procedures imposed on recipients of delegated authority (McCubbins 1985; McCubbins *et al.* 1987, 1989; McNollgast 1994; Epstein & O’Halloran 1999) or in policy details specified in

legislative language (Huber & Shipan 2002). Discretion, then, is delegation net of constraints.

In their work, Bolton & Thrower (2019) construct a discretion measure by dividing the new budget authority for an agency in each year (serving as a proxy for delegation) by the length of the report from the appropriations subcommittee with jurisdiction over the agency (serving as a proxy for constraints).¹ The logic of this measure builds on the work of Huber & Shipan (2002) but explicitly accounts for the possibility that longer texts (whether committee reports or legislation) may be associated with more delegation—and not solely more constraints (Bolton & Thrower 2019, pg. 1271).²

In this paper, we eschew the concept of discretion and focus on delegation. Our reason for this focus is that delegation—the decision to assign a task to an executive branch actor—can be directly observed in legislative text. Discretion, on the other hand, while undeniably a vital concept, does not readily admit itself to clean observation or measurement. This measurement issue can be illustrated through two critiques: one about units of measurement and the other about irreducible ex-ante uncertainty.

Whereas delegation can be measured as a dichotomous variable based on its presence or absence in legislative text, the scale on which discretion should be measured is unclear.³ We do not know what a unit of discretion is because we are not clear in empirical work about the dimensions of policy space within which executive branch actors operate. Whereas legislative constraints can be directly observed in legislative text and are theoretically connected to the concept of discretion, it is unclear how much any given constraint restricts discretion.

The second critique concerns ex-ante uncertainty about the scope of discretion that

¹For more specifics on this measurement strategy, see Bolton & Thrower (2019, pgs. 1268-1270).

²Lowande & Shipan (2021) take a different approach to measuring discretion. Focusing specifically on the discretion that presidents have over different policy areas, the authors conduct a survey in which subject area experts are given pairwise comparisons of randomized policy areas and asked to select the area over which the president has more discretion. They then estimate a random utility model on the survey responses to recover a latent measure of discretion.

³Of course, measuring delegation dichotomously evades this scaling problem and implicitly assumes that all delegations are equally important. We agree with McCann & Shipan (2022) that, while differentiating between the degrees of importance associated with different delegations of authority constitutes an important question, we must first be able to accurately identify delegation in legislation. Our work here allows us to do that across a wide array of bills.

exists when a law is enacted or a committee report is issued. While delegation is written directly into legislative documents — and the discretion associated with that delegation is shaped by the text of reports, delegating statutes, and other relevant laws like the Administrative Procedure Act — it is not possible to know ex-ante the true scope of discretion allowed to an executive branch agency by any given law. This is revealed over time as agencies make rules and regulations pursuant to delegated power and as courts determine the legality of agency actions. Because there is considerable uncertainty surrounding both the actions that will be taken in response to any legislative text and how those actions will be received by other relevant actors (courts, future legislative coalitions, plaintiffs, etc.), we are hesitant to assign a fine-grained, text-derived measurement to the concept of discretion.

The issue of student loan debt relief during the COVID-19 pandemic further illustrates the problems of ex-ante uncertainty associated with attempting to measure discretion. The statutory basis for the pause placed on student loan debt collection during the Trump administration—and the attempted forgiveness of that debt during the Biden administration—was the Higher Education Relief Opportunities for Students (HEROES) Act of 2003. This law gave the Secretary of Education the authority, in the case of a war or national emergency, to “waive or modify” any statutory or regulatory provision associated with student financial assistance programs. How much discretion is associated with this delegation of authority, and how should we measure it? For one thing, it is discretion that exists only under certain conditions (wars, national emergencies). For another, it is impossible to predict how present or future Secretaries of Education will use that discretion when it is conditionally activated. Further, it is impossible to predict how courts will react. From what measure of discretion could we have known that this delegation would have been responsible for an estimated cost of \$100 billion to the federal government, an attempt to cancel student loan debt collection entirely for a large swath of borrowers, and a Supreme Court case nullifying that attempt? Even knowing all of this in retrospect, what value of discretion would we assign to this particular part of the HEROES Act when it passed in 2003?

For the reasons articulated above, our paper focuses exclusively on the observation and measurement of delegation. The following section describes our methodology for identifying delegation in legislative texts. As mentioned above, delegation is conceptually straightforward and relatively easily identifiable for a reader familiar with legislative texts. However, the difficulty is that this time-intensive reading approach does not readily scale up for large-N empirical research. Our method provides a scalable solution to coding legislative texts for delegatory language.

An Active Learning Convolutional Neural Network for Classifying Text

Classification is one of the most popular objectives in all text-as-data work. Generally falling under the umbrella of supervised learning, classification tasks take a set of inputs, along with their corresponding outputs (also called labels), with the goal of discovering some underlying map between input and output. Supervised methods often require thousands of training examples, rendering them a non-starter for many researchers and projects. However, there are often creative ways to reduce the effort required. We see an example related to our own in Anastasopoulos & Bertelli (2020), who use existing classifications of agency delegation in EU legislation to perform supervised labeling of other years not covered in the dataset.

Active learning is a machine learning approach to bolster classification performance by selecting (or “querying”) further examples to use in model training. This querying allows an algorithm to obtain higher classification accuracies as training data are added than if new examples were instead randomly chosen. An active learner typically poses its queries over unlabeled data, which are then labeled by a human annotator and subsequently added to the dataset. Active learning is well-motivated in many modern machine learning problems, where unlabeled data may be abundant, but labels are complicated, time-consuming, or expensive to obtain (Settles 2009; Miller *et al.* 2019).

Researchers may find that they need to frequently retrain or update their models, particularly given new information they discover in the course of their exploration. Active learning offers a judicious way to update with new examples. We provide a motivating example in identifying delegation to administrative agencies. The general framework for delegating authority is relatively straightforward: some agency is given a task — often told it “shall” or “must” do something. A standard learner would identify the agency named and the verbs specifying the task and then use that information to identify most instances of delegation. However, there are hundreds of currently active agencies and programs, some of which have unusual names (the “Corporation for National and Community Service” as an example), that would reduce the likelihood of delegation being properly identified.⁴

We encountered this problem early on when discussions of the “Attorney General” were frequently mislabeled because, up until that point, there were no observations in our training data with a cabinet-level secretary referred to as anything but “Secretary.” Because the specific words “Attorney General” have few analogous positions in other departments, it would have to be hand-coded explicitly for the model to learn what it is. Generally, this is not a complicated fix: hand label some of these aberrant observations, and it should solve the problem. However, what we could not know *a priori* was what exact issues were going to appear: moving to an interactive labeling and machine-learning framework alleviated these concerns because the classifier would be able to tell us where it was having difficulties discriminating between classes. Learning on the go was the preferred option because, though our classification scheme is simple, there are enough moving parts that it is hard to know precisely what problems would have arisen before we started the coding.⁵

In a typical supervised machine learning environment, a model is trained on a training partition of the data and evaluated on a testing partition. Given the performance, applied

⁴Unless that agency also showed up in the training data which, given that there are hundreds of agencies, is likely to miss many.

⁵This is a limitation of grammar-based approaches to similar problems. Since our method adapts and learns as we hand-code more and more uncertain observations, it begins to cover the realm of possible mismatches more broadly. Given the complexity of congressional language and the continuing evolution of the legislative agenda, we have concerns, though see Vannoni *et al.* (2021) for counter examples.

researchers may then decide to “deploy” the model for some practical purpose: for instance, the automated labeling of new examples. They may also determine that the model performance, as evaluated on the test data, was insufficient, and so they may choose to label more examples for model training. For our case, we argue that active learning should be used when selecting new examples to label, with the criterion for selection being the model’s uncertainty in labeling new examples. After querying these uncertainly labeled documents from the model, they may be manually labeled by the researchers and appended to the training dataset. The process of training, evaluation, and querying then repeats until model evaluation suggests the model is robust enough to be deployed to label all remaining examples automatically.

As mentioned above, our criterion for actively querying new data examples is the model’s uncertainty in label prediction — “uncertainty sampling.” In the case of classification, this strategy amounts to identifying the observations closest to the classification boundary, such as examples with the smallest margin in max-margin models (e.g., SVM) or examples with logits closest to 0.5 for logistic regression.⁶ For our problem, we have implemented uncertainty sampling, as it is the most widely used, computationally straightforward, and flexible for comparison between various methods (Tong & Koller 2001; Settles 2009).

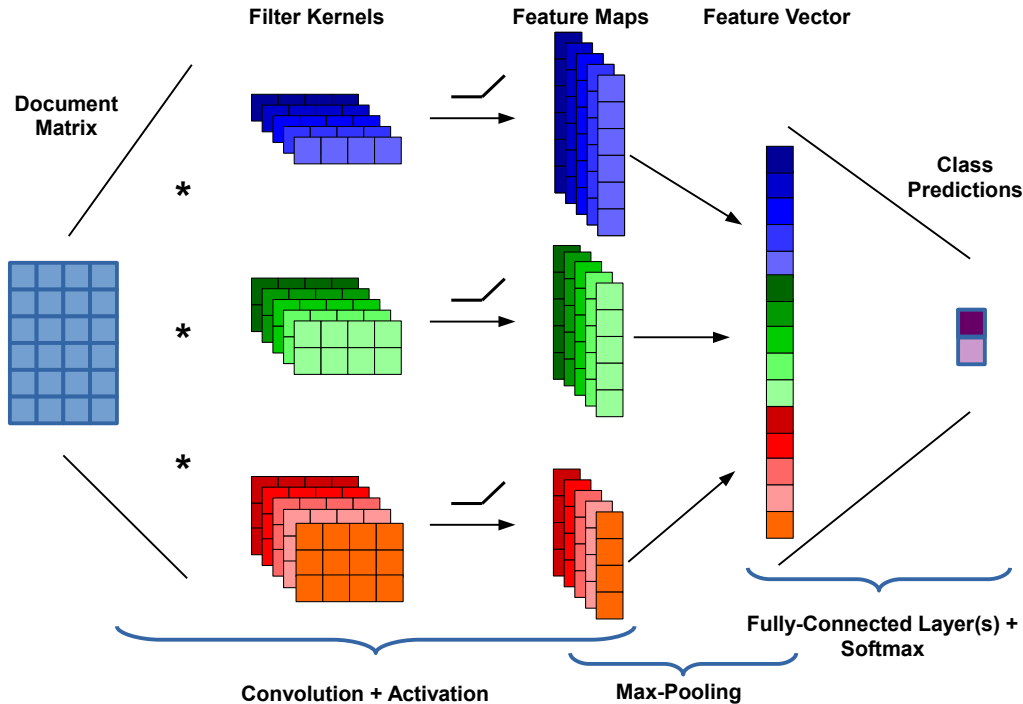
This paper uses a convolutional neural network (CNN) with a multi-layer perceptron (MLP) as our primary classifier. We note that a variety of other supervised learning methods can accommodate active learning, and the choice of model is the researcher’s preference. We use uncertainty sampling, querying those examples with logits closest to 0.5.

Convolutional Neural Networks for Text Classification

We follow the example of Kim (2014) who defined a CNN for text classification, as well as Zhang & Wallace (2015) who provide practical guidance on using such models. The primary

⁶Other common querying strategies are ensemble approaches — wherein queries are made by identifying observations for which there is classification disagreement between multiple models — and expected-model-change approaches, in which examples are selected that would most significantly change the current model (Settles 2009).

Figure 1: Illustration of CNN for text classification



For this example, $d = 4$, there are 5 filters of lengths $\{1,2,3\}$, and there are 2 classes.

advantages of using deep learning for our text analysis — rather than “bags of words” approaches (Grimmer & Stewart 2013) — are to model the similarity between words and to account for word order in text sequences. Both of these advantages involve considering words in context. Underpinning the use of CNNs for text analysis is the distributed representation of words, wherein each word in a vocabulary is associated with a real-valued feature vector (Mikolov *et al.* 2013; Pennington *et al.* 2014). These expressive vector representations encode many linguistic regularities and patterns, such as the relationships between synonymous words, and their use has been shown to improve the accuracy of supervised NLP tasks (Turian *et al.* 2010). Unsupervised training of word embeddings is typically accomplished by predicting the incidence of words given local context words, but practitioners may wish to instead randomly initialize word embeddings to be fully learned as parameters of their specific task (Rodriguez & Spirling 2021). While we opt for the latter strategy, we include

in the Appendix a visualization of word embeddings learned via `word2vec` (Mikolov *et al.* 2013) on the corpus of bills from the 110th Congress.

Figure 1 visually illustrates how our CNN takes our text data and generates our predictive labels.⁷ This example is included for illustrative purposes only; for further details, please view Appendix Section 2, which details exactly how our CNNs were fit. All of our text classification models were fit in TensorFlow, taking advantage of its robust deep learning framework.

Data

We utilize data for all versions of all bills (both successful and unsuccessful) from the 110th and 111th Congresses. We separate each version of each bill into titles and analyze them at the bill section level. We do this for three reasons: first, because each title in each bill deals with a particular agency or activity and likely contains an entire delegatory phrase, keeping the task more straightforward. Second, any given bill could delegate authority to multiple agencies in multiple titles, so to avoid missing any additional delegations, we wanted to reduce it to units that are about a single delegation. The final reason we chose bill titles is that bill titles are the smallest comparable distinct units of a bill: titles are more comparable to one another than either a sentence or an entire bill would be.⁸

Our complete dataset has several components. The division of our dataset is into labeled and unlabeled sub-datasets. The labeled component comprises 2098 bill sections from the

⁷To prepare the text of our dataset for the model we have described, we performed the following pre-processing steps: converted all text to lowercase, removed extraneous whitespace and punctuation except periods and commas, and replaced all digits with a "digit" token. Then, each word token was assigned a unique integer identifier, in essence creating a dictionary between words and corresponding integers. This dictionary was used to map the words of each bill section to a sequence of integers, and these sequences of integers were used as the inputs to our neural network model.

⁸While it might seem advantageous to go more fine-grained than the bill section, there are fundamental limitations in how bills are written that make this problematic. For example, the most common unit of analysis in natural language processing tasks is the sentence. However, the construction of any given sentence within a bill is contingent primarily on the remaining information presented at the title level. Studying bills at the title level will ultimately let us make better inferences about the agencies or programs to which Congress has delegated.

110th Congress that were read by human annotators and assigned a binary label with respect to delegation; the unlabeled component contains the remaining 137,616 bill sections from the 110th Congress as well as all sections from the 111th Congress. As is standard practice within machine learning, we divided our labeled data into subsets for training, validation, and test. In performing this splitting, we ensured that all titles/versions from the same bill were apportioned to the same subset. We randomly divided the bills in our labeled data into training, validation, and test using proportions of 65%, 15%, and 20%, respectively. The remaining bills from the 110th Congress were also apportioned into training, validation, and test using the same proportions. The numbers of bills (by bill number) and examples (e.g., bill section) for each subset are presented in the Appendix.

Delegation Coding

Essential to our project is a consistent definition of delegation to administrative agencies. An act of delegation is a mandate or permission for a federal agency or program (including the president) to exercise public authority in some way (see McCubbins *et al.* 1987; Kiewiet & McCubbins 1991; Huber & Shipan 2002; Gailmard & Patty 2012a, for a discussion of this point). For our task, we stated that allocating money for federal agencies to spend, instructing agencies to promulgate rules, granting agencies the ability to exempt themselves from preexisting rules, requiring agencies to compile reports or commission pilot studies, and charging agencies with the enforcement of specific policies are all delegation (Kiewiet & McCubbins 1991).

For the hand-coding, we gave straightforward instructions on identifying delegation. First, is Congress acting upon an administrative agency? This will include all references to both the agency itself and the person in charge of that agency. We operated with a list of administrative agencies and matched each instance of delegation to one of those agencies.⁹

⁹Most often, if Congress is referring to a governmental entity (except for organizations already within Congress, which they make apparent), it is an administrative agency. We make an exception for delegating to the courts, states, and local governments because those tasks are defined differently.

Second, what is the title asking the agency to do? In general, Congress delegates authority by asking an agency to perform a specific task, collect information, write new regulations, hire people, write a report to Congress on their activities, delegate authority to sub-agencies or outside of government, and make or distribute an award, among many other things. A bill is not delegating authority if it only appropriates money, if they are referencing actions already taken, or if Congress is writing new rules or regulations. Keeping these actions separate allows us to track statutorily derived authority for the agencies, not merely what funds they have been allotted.

Below are example bill titles that the active learner selected in early runs as uncertain and how they were coded.

- Section.2402. energy conservation projects. using amounts appropriated pursuant to the authorization of appropriations in section.2403.a.6, the secretary of defense may carry out energy conservation projects under chapter 173 of title 10, united states code, in the amount of 800000 (delegates authority to an administrative agency)
- Section.2. reemployment of foreign service annuitants... the authority of the secretary to waive the application of subsections a through d for an annuitant pursuant to subparagraph c of paragraph 1 shall terminate on September 30, 2008. the authority of the secretary to waive the application of subsections a through d for an annuitant pursuant to subparagraph c ii of paragraph 1 shall terminate on September 30, 2009 (does NOT delegate authority to an administrative agency)

In the above examples, it is clear why the algorithm would have selected them as ambiguous classifications and why human readers would classify them correctly. Take the top section, dealing with “Energy Conservation Projects.” A quick read of the section makes it clear that Congress is delegating authority to Defense (through the Secretary of Defense) to spend \$800,000 on energy conservation projects. The classifier may have had issues with the added verbiage of the task (“may carry out”) and the addition of US code language in between. In the second section, where Congress is setting up “foreign service annuitants”,

it is clear that the agency is not being given an extra task or authority but only describes how applications must be processed (and when they terminate). This section is an example where the language (context-free) would indicate the possibility of delegating authority, but the additional context (plus a close reading) makes it clear that this is not occurring. These are only two examples pulled from an early run of the active learning module, set up to illustrate the nature of the classification task.¹⁰

Classification and Delegation Results

We explore classification modeling results in the Appendix. To summarize, To evaluate active learning performance, we started with a small training set and iteratively added new examples, either through active learning or random sampling, and then measured validation performance. Our results showed that active learning improved classification accuracy quicker than random sampling, particularly for training sizes of 20-800 examples for the SVM and 200-600 examples for the CNN. We also benchmarked overall classification performance against more conventional ML approaches. We found that the CNN outperformed the SVM and other tf-idf baselines. See Appendix Section 2 for details of the model, Section 3 for comparisons and how classifier performance improved with active learning versus conventional random sampling, and Section 4 for details of different baseline specifications for the ML models.

With delegation predictions for each version of each bill section from the 110th and 111th Congress, the remainder of the paper will examine the consequences of these delegation classifications. First, we present descriptive data on the distribution of delegation in introduced bills across cabinet-level agencies, broken down by sponsor party. Then, we use our delegation data to evaluate aspects of the ally principle. Here, we are particularly in-

¹⁰A concern with our approach is that the data we generate from our models is only as good as our human coding. We did manual evaluation and validation of the hand-coding that made up the training set and we've included the coding instructions in Appendix 8. The entire training set with the hand labels will be provided with replication data for interested readers to validate themselves.

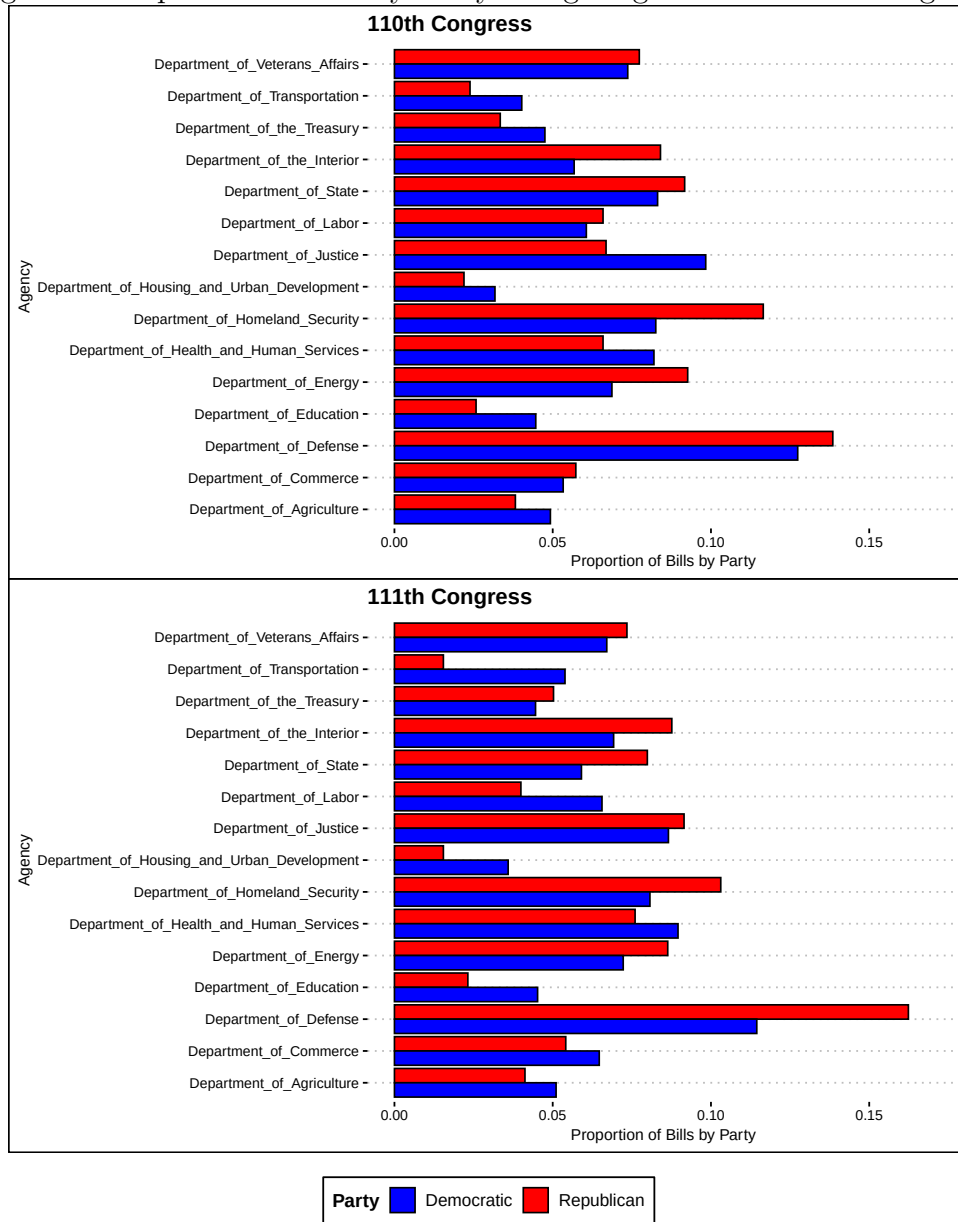
terested in testing whether congressional delegation decisions are more responsive to stable agency characteristics—like ideology and structural independence—or to shifts from divided to unified government. Our results suggest that congressional decisions about delegation are shaped by the stable contours of the administrative state rather than the top-level political leadership that changes with presidential administrations.

What Agencies Get Delegated To?

In this section, we attempt to discern the delegatory approach of Congress descriptively by observing the delegatory patterns in our new dataset. The transition from the Bush Administration to the Obama administration denotes a clear opportunity to view the preferences of Congress remaining stable (Democratic House and Senate) while the White House changes partisan affiliation. If there are substantial shifts in delegation in divided versus unified government, this would be strong evidence that MCs are treating the agencies as the sum of their political appointments; if there are minimal or no changes, this would be evidence that MCs treat the agencies as stable reflections of the bureaucracy, and are more consistent with the administrative state (Lowande 2018).

Figure 2 shows the proportion of bills delegating to cabinet-level agencies by sponsor party for the 110th and 111th Congresses respectively. Within each Congress, there are a few partisan patterns that could feasibly be driven by ideological considerations. For example, Republicans tend to sponsor more bills delegating to more conservative departments, such as the Department of Defense and the Department of Homeland Security. Democrats, on the other hand, sponsor more bills that delegate authority to more liberal departments, such as the Department of Education. However, most of the partisan differences in delegating activity are not particularly large. Additionally, this descriptive look at the data does not suggest large shifts in delegating behavior brought about by a change in presidential administration. The patterns are mostly stable in the wake of the transition from the Bush to the Obama presidency, and some of the cross-Congress changes that do occur are contrary to the

Figure 2: Proportion of Bills by Party Delegating to Cabinet Level Agencies



ally principle. For example, there are relatively more Republican-sponsored bills delegating authority to the Department of Justice under the Obama administration than under the last Congress of the Bush administration.

Theoretical Expectations for Variation in Delegation

Having described our method and presented descriptive data on delegation during the 110th and 111th Congresses, we move on to using our measure to test an important theoretical proposition from the literature. Perhaps the most prominent finding from the extensive literature on delegation is the ally principle, which posits that interbranch ideological conflict drives variation in legislative delegation decisions. Specifically, the ally principle yields a directional hypothesis that the scope and amount of delegation decreases as the ideological distance between the legislature and the recipient of delegated authority—usually an executive agency—increases.

The ally principle rests on a straightforward assumption that legislative principals are averse to losing policy utility at the hands of unfaithful agents. While some literature has articulated compelling theoretical reasons the ally principle may be violated in practice (Huber & McCarty 2004; Gailmard & Patty 2012b; Miller & Whitford 2016), the body of empirical literature confirms that, on average, the principle holds (Epstein & O’Halloran 1999; Huber & Shipan 2002; Clinton *et al.* 2012; Farhang & Yaver 2016; Bolton & Thrower 2019). Our new measure of delegation allows us to add to this literature in two specific ways that both shed light on previously unappreciated facets of the ally principle.

First, our methodology enables us to identify delegatory language in large datasets of legislation spanning multiple Congresses. This allows us to expand our focus beyond a small set of laws to the universe of all legislation within a certain time period. This broader scope gives us the ability to test the generalizability of the ally principle across different legislative contexts. For example, we are able to determine whether findings supporting the ally principle are artifacts of a narrow focus on a small subset of important laws, or if these trends are more broadly applicable. There is a growing body of literature that shows that Congress treats “significant” legislation differently than ordinary legislation, and most of the work on the ally principle has focused on high-profile or otherwise seemingly important legislation (Farhang & Yaver 2016; Shaffer 2020).

Secondly, the dataset we create—which identifies every instance of delegation, as well as the recipient of that delegation, in all bills introduced in the 110th and 111th Congresses—allows us to gain leverage on fundamental questions about how legislators view the executive branch. Any test of the ally principle requires ideological measures for both the principal and the agent—and a basic prerequisite for this measurement strategy is to be able to specifically identify an agent in each bill. However, this identification is not necessarily a straightforward task. If the president is able to set the ideal point of each executive agency—an assumption common in much of the delegation literature—a simple indicator for divided government may be sufficient for testing the ally principle. If, on the other hand, agency ideology is relatively stable across presidential administrations but varies substantially across agencies, agency-specific ideology measures may be necessary.

Our data uniquely equip us to differentiate between two potential legislative views of the executive branch—one as a presidentially-controlled unitary institution, and the other as a complex and internally-differentiated conglomeration. Because our data span the 110th and 111th Congresses, we are able to observe variation in regime type (from divided to unified government) while keeping the partisanship of the congressional majority constant. Using agency ideology and independence measures developed by other scholars, we are also able to measure the cross-agency variation that exists within the executive branch at any given point in time.¹¹ Importantly, our measurement strategy here varies from Clinton *et al.* (2012), who assign a primary recipient of delegated authority for each bill. As described in more detail below, we create bill-level measures that make use of information on each agency receiving delegated authority—an advantage that we have because of our delegation data.¹²

¹¹While these agency characteristics are themselves products of interbranch (and intrabranh) political struggles (Moe 1989; Lewis 2004), they continue to affect subsequent delegation decisions long after the statutory establishment of the agency in question.

¹²Over 55% of the bills in our data delegate to only one agency, making agency-level information completely consonant with bill-level information for those observations. To check if our findings are robust to different treatments of the remaining bills—those that delegate to multiple agencies—we present a number of different measurement strategies in the Appendix. In Appendix 9.3, we fit models in which we match each bill to the agency that is the recipient of the largest number of delegations in that bill, similar to the strategy used in Clinton *et al.* (2012, pg. 350). In those models, each bill is assigned the ideological and independence characteristics of the agency to which it delegates the most. In Appendix 9.4, we fit models

Models

Our intent in this section is to construct a set of empirical tests to evaluate scope conditions of the ally principle. Much of the literature on delegation operationalizes the ally principle by comparing instances of unified government with instances of divided government (Epstein & O'Halloran 1996, 1999; Huber & Shipan 2002; Farhang & Yaver 2016), implying that it is the partisanship of the president that drives variation in congressional delegation decisions. However, given the diversity of executive branch agencies—both from an ideological and a structural standpoint—it is likely that the effect of regime type on congressional delegation decisions is conditioned by specific agency characteristics. New developments in the measurement of two such agency characteristics, ideology (Clinton & Lewis 2008; Clinton *et al.* 2012; Chen & Johnson 2015) and independence (Selin 2015), allow us to test this proposition.

Agency ideology refers to the aggregated policy preferences of each executive branch agency. We use a measure developed by Clinton *et al.* (2012), who administered a survey to agency appointees and careerists asking them to take positions on salient congressional roll call votes. This methodology yields agency-level ideological scores that are on roughly the same scale as the ideological scores for members of Congress.

Agency independence generally refers to the extent to which executive branch agencies are insulated from political pressure exerted by Congress or the president. To operationalize this concept, we use measures developed by Selin (2015), who estimated agency ideology on two dimensions—one tapping limits on the appointment of key agency officials and the other tapping the extent to which agency decisions are subject to political review.

The structure of our data—which are at the bill level and include variables measuring the number of delegations of authority to each agency—necessitates some transformation of these agency-level ideology and independence measures. For the ideology measure and both independence measures, we create bill-level weighted averages using information on all

on only the subset of bills that actually delegate to a single agency. Results from both of these appendices are substantively similar to those presented below.

agencies receiving delegated authority in each bill. Each agency has an ideology score from Clinton *et al.* (2012) and two independence scores from Selin (2015), and for each bill, these scores are multiplied by the number of individual delegations of authority to that agency in that bill. The resulting products, which are ideology and independence scores weighted by the prevalence of each agency in each bill, are summed across all agencies and then divided by the total number of delegations in each bill to create bill-level agency ideology and agency independence “profiles.”¹³ The following equation is applied to each bill to arrive at these profile scores:

$$ideo_j = \frac{\sum_{i=1}^n \theta_i d_{ij}}{\sum_{i=1}^n d_{ij}}$$

$$ind_j = \frac{\sum_{i=1}^n a_i d_{ij}}{\sum_{i=1}^n d_{ij}},$$

in which the subscript j indexes bills, the subscript i indexes agencies, θ_i is an agency-specific ideal point, and d_{ij} is an agency-bill-specific count of delegations. For independence, we use the same formulation, but with agency-specific independence scores (a_i) rather than agency-specific ideal points θ_i .

To test for the conditions under which the ally principle operates through presidential partisanship, we fit a series of regressions. For our dependent variable, we use a bill’s delegation ratio as defined by Epstein & O’Halloran (1999). The delegation ratio Δ_j for bill j represents the number of bill sections delegating authority d_j to administrative agencies divided by the total number of sections in the bill p_j :

$$\Delta_j = \frac{\sum_{j=1}^n d_j}{\sum_{j=1}^n p_j}$$

Whereas Epstein & O’Halloran (1999) create their delegation ratios for a select number

¹³Appendix 9.3 includes models fit with “unweighted” agency characteristics, which simply take the average ideological and independence scores across all agencies that receive delegated authority in a given bill without regard to the relative prevalence of each agency in the bill. The results from the models fit with these “unweighted” variables are substantively similar to those reported below.

of laws based on the coding of CQ summaries, our method allows us to create delegation ratios for a broader universe of bills using entire bill texts. Our use of the delegation ratio as a dependent variable in tests of the ally principle is based on the assumption that bills with high delegation ratios delegate substantively more authority, while bills with low delegation ratios delegate substantively less authority. This fits well with the discussion in Epstein & O’Halloran (1999, ppg. 94 & 95), who find that laws in their dataset with low delegation ratios “delegated very little authority to the executive,” while laws with high delegation ratios “granted the executive great leeway to set policy.”

We fit a series of regressions predicting a bill’s delegation ratio Δ_j based on the distance between bill-level ideology and the relative majority party medians, two dimensions of agency independence, and a dummy variable for unified government.¹⁴ Because the delegation ratio is constrained between zero and one, we use a beta regression to fit the model.¹⁵

Results

The models presented in Table 2 below are fit on six different subsets of data—a subset of all HR bills introduced, a subset of all HR bills that passed the House, a subset of all S bills introduced, a subset of all S bills that passed the Senate, a subset of bills that were enacted into law, and finally a subset of important bills by Mayhew’s classification (Mayhew 1991).¹⁶ This setup allows us to observe variation in the effects of agency ideology and independence on delegation ratios across different legislative contexts. Models fit on all introduced bills will include messaging bills not intended to become law, so evidence of the ally principle may be diluted in these larger samples. Models fit on bills that passed either chamber, or that became law, yield estimates of the effects of agency characteristics on delegation in actual

¹⁴Some models also include fixed effects for policy area and for bill version. More detail on these measures is found in the subsequent section.

¹⁵Similar results are found if we fit our models with OLS. See Appendix 9.2 for details.

¹⁶The models are fit on the subset of bills that delegate authority to actors in the executive branch—bills that do not include any delegation are excluded. Therefore, what is being analyzed is variation in the *amount* of delegation in each bill (measured by the delegation ratio), as opposed to the initial decision of whether or not to delegate.

lawmaking.¹⁷

Following the logic of Cox & McCubbins (2005), we assume all bills that pass a legislative chamber have the support of the majority party in that chamber, which for our time period (110th and 111th Congresses) is the Democratic party in both the House and the Senate. Therefore, the coefficients on the agency ideological profile variable can be interpreted as the expected change in delegation to an agency as that agency becomes increasingly ideologically distant from the Democratic congressional majority.¹⁸ Similarly, the coefficients on both agency independence profile variables capture the expected effect of increasing agency independence on delegation.

The dichotomous indicator for unified government straightforwardly tests the main expectation of the ally principle as traditionally operationalized. In our specific case, coefficients on this variable can be interpreted as the change in delegating activity from a Democratic Congress attributable to the change in the presidency from George W. Bush to Barack Obama, the same change exploited by Lowande (2018) to test congressional relations with the changing executive branch.

Because delegation has been shown to vary across issue area, all models presented below, with the exception of the model fit on the Mayhew bills subset in Table 2, include fixed effects for the Policy Agendas Project topic area of each bill, as coded by the Congressional Bills Project (Adler & Wilkerson 2011). Because the unit of analysis for the data we use is bill-version, we also include a fixed effect for bill version (introduced, reported from committee,

¹⁷Appendix 9.4 includes models fit on subsets of the data based on measures of substantive importance. The results presented from those models are substantively similar to those presented in the paper, demonstrating that the main findings are not driven by the inclusion of commemorative legislation of little substantive importance.

¹⁸The ideology variables are constructed as the absolute difference between the ideological score of the median Democrat in the relevant chamber and the bill-level agency ideological profile score (which is the weighted average of the ideological score of each agency delegated to in each bill). For our models fit on bills that passed the House, the score is constructed using the median House Democrat in each Congress. For the models fit on bills that became law, we take the median ideological score of House and Senate Democrats pooled together by Congress. Models in Appendix 9.3 use two differently-constructed ideological measures, and the results from those models demonstrate that the findings presented here are robust to different assumptions. One set of models uses the raw agency ideological scores (without subtracting them from any legislator’s ideological score), and another uses the absolute distance between the bill sponsor’s DW-NOMINATE score and the agency’s ideological score.

passed House, etc.) to account for any systematic variation in delegatory scope that may occur as bills advance through the legislative process. These fixed effects are included in all models below, with the exception of models that focus exclusively on enacted bills —because these models are only fit on data containing the enacted versions of bills that became law.

Table 2: Estimating Delegation Ratio as a Function of Regime Type and Agency Independence and Ideology

	<i>Dependent variable:</i>					
	Delegation Ratio					
	HR Bills	HR Bills (Passed House)	S Bills	S Bills (Passed Senate)	Laws	Mayhew
Unified Government	0.010 (0.022)	−0.030 (0.031)	0.026 (0.031)	0.162* (0.076)	−0.087 (0.080)	−0.312 (0.331)
Agency Ideol. − H.Maj. Median	−0.051* (0.025)	−0.170* (0.037)				
Agency Ideol. − S.Maj. Median			−0.068* (0.032)	−0.368* (0.080)		
Agency Ideol. − C.Maj. Median					−0.455* (0.100)	−0.545 (0.320)
Agency Independence Profile D1	−0.340* (0.023)	−0.352* (0.033)	−0.325* (0.034)	−0.135 (0.073)	−0.236* (0.081)	−0.981 (0.555)
Agency Independence Profile D2	−0.054* (0.022)	−0.147* (0.034)	−0.114* (0.030)	−0.528* (0.080)	−0.344* (0.095)	0.258 (0.496)
Num.Obs.	8520	4074	4301	775	492	24
R2	0.337	0.512	0.213	0.432	0.586	0.405
Topic Area FEs	✓	✓	✓	✓	✓	
Bill Version FEs	✓	✓	✓	✓		

* $p < 0.05$

Note: The three ideological distance variables in the table are bill-level variables measuring the absolute difference of the DW-NOMINATE score of the median majority party member in each chamber (or in both chambers combined, as is the case for the model fit on bills that became law) and the weighted average of the Clinton *et al.* (2012) agency ideology scores for agencies receiving delegated authority in each bill.

The results presented in Table 2 demonstrate that individual agency characteristics —specifically agency ideology and independence—retain statistically significant explanatory power even when conditioning on the existence of unified government.¹⁹ In the model fit on bills that passed the Senate —the only model in which the coefficient on the unified government indicator is statistically significant—the coefficient is in the positive direction predicted by the ally principle. However, all other models have a null result for unified government.

The findings on agency ideology and independence are mostly consistent across all

¹⁹In Appendix 9.1, we demonstrate that the mostly null results with respect to the unified government variable are robust to a simple model structure that regresses delegation ratio on a dichotomous indicator for unified government, without agency-specific variables.

models besides the one fit on the subset of bills classified as important by the Mayhew coding system (Mayhew 1991), which is underpowered given the small number of observations. The overall trend demonstrates that the Democratic majorities of the 110th and 111th Congresses wrote less delegation into bills that delegated to more conservative and more independent agencies, regardless of whether these bills were passed during unified or divided government. The results presented in Table 2 suggest that, when making decisions about the scope and extent of delegation, congressional majorities are more attuned to the relatively stable traits of agency ideology and independence than they are to changes in political leadership in the White House and at the top levels of agency hierarchies.²⁰

The models presented in Table 3 test our expectation that the effect of a change in presidential administration on congressional delegation activities should be conditional on the level of agency independence. Our expectation is that delegation to agencies that are designed by statute to be particularly susceptible to political influence (agencies that would score low on Selin’s independence measures) will be more affected by changes in presidential administration. Specifically, we include an interaction between the second dimension of the independence measure—which measures independence from policy review by political principals—with the unified government indicator. Here, we are using the independence measure as a proxy for the responsiveness of agency policymaking to the president and politically-appointed leadership, and seeking to test whether the effect of that political responsiveness on delegation is conditional on unified or divided government.

It also may be the case that agency ideology affects congressional delegation decisions differently at different levels of agency independence. We include an interaction between the second dimension of the independence measure and the ideological distance measure to see if

²⁰It is worth noting here that the agency-level ideology and independence measures we use to construct our variables are time-invariant, which assumes that these constructs are not meaningfully affected by changes between unified and divided government. Models included in Appendix 9.3 relax this assumption by using time-varying agency ideology scores from Chen & Johnson (2015). The results from these models, which employ agency fixed effects and an indicator variable for unified government in the 111th Congress to isolate the effect of within-agency ideological from one administration to the next, largely support the findings presented here. Most variation in delegating activity seems to be driven by ideological differences across agencies, rather than ideological shifts within agencies over time.

Table 3: Interaction Models

	<i>Dependent variable:</i>		
	Delegation Ratio		
	HR Bills (Passed House)	S Bills (Passed Senate)	Became Law
Unified Government	−0.011 (0.039)	0.448* (0.095)	−0.004 (0.109)
Agency Independence Profile D1	−0.317* (0.033)	−0.171* (0.072)	−0.227* (0.081)
Agency Independence Profile D2	−0.318* (0.053)	−0.272* (0.110)	−0.398* (0.147)
Unified Gov. X Agency Ind. D2	−0.015 (0.037)	−0.503* (0.101)	−0.101 (0.094)
Agency Ideol. − H.Maj. Median	−0.224* (0.039)		
Agency Ideol. − S.Maj. Median		−0.394* (0.082)	
Agency Ideol. − C.Maj. Median			−0.477* (0.103)
Agency Ideol. − H.Maj. Median X Agency Ind. D2	0.282* (0.059)		
Agency Ideol. − S.Maj. Median X Agency Ind. D2		0.059 (0.154)	
Agency Ideol. − C.Maj. Median X Agency Ind. D2			0.238 (0.200)
Num.Obs.	4074	775	492
R2	0.517	0.452	0.597
Topic Area FEs	✓	✓	✓
Bill Versions FEs	✓	✓	

* p < 0.05

Note: The three ideological distance variables in the table are bill-level variables measuring the absolute difference of the DW-NOMINATE score of the median majority party member in each chamber (or in both chambers combined, as is the case for the model fit on bills that became law) and the weighted average of the Clinton *et al.* (2012) agency ideology scores for agencies receiving delegated authority in each bill.

the ally principle expectation that delegation will increase with ideological proximity holds across different types of agencies. To narrow our analyses here to more serious attempts at lawmaking, we present only the models fit on subsets of bills that passed the House (first

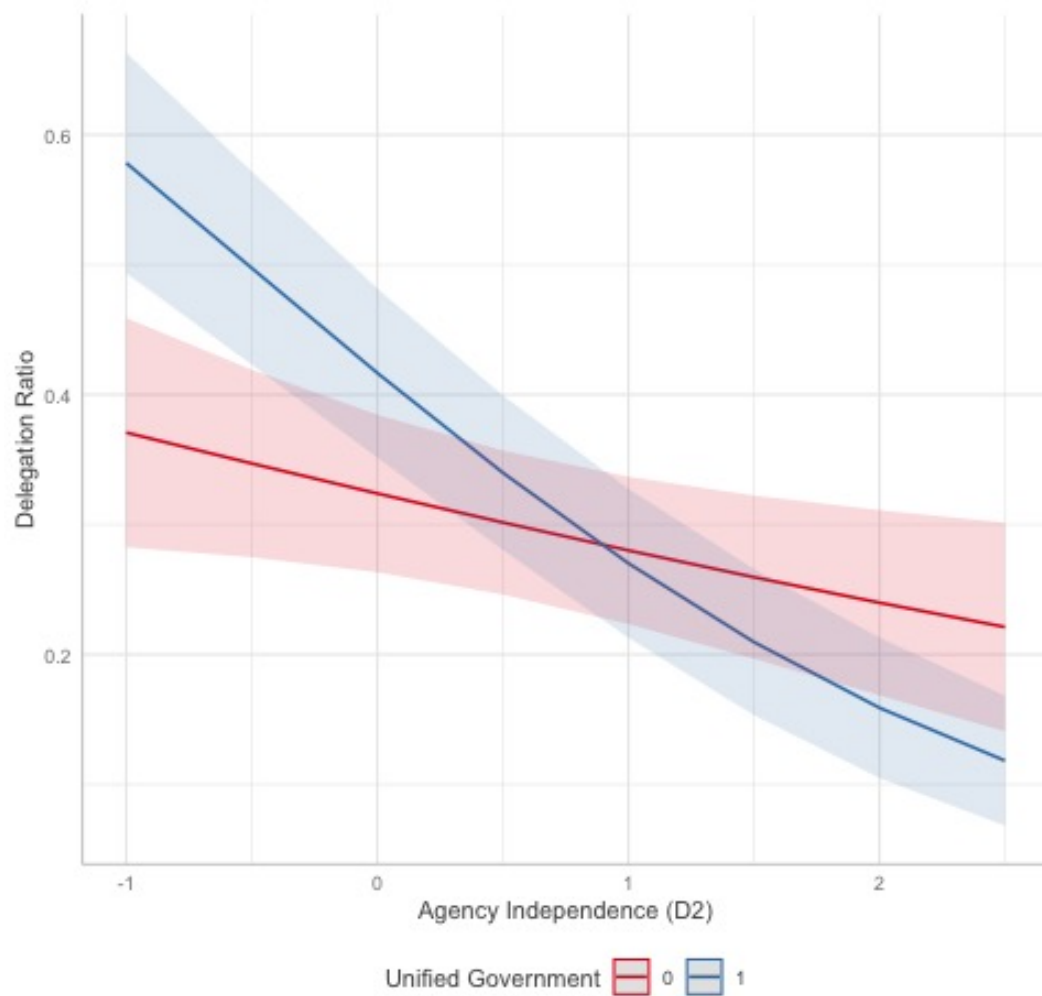
model), passed the Senate (second model), and became law (third model).

The coefficient on the first interaction, between unified government and agency independence, is negative in all models, and statistically significant in the model fit on bills that passed the Senate. We explore this finding more directly in Figure 3, which plots the marginal mean effects from the second model from Table 3. In this plot, the X-axis is the second dimension from Selin’s independence measure, with larger values corresponding to more independent agencies, and the Y-axis is the estimated marginal Delegation Ratio for the bill. We differentiate between unified and divided government. Consistent with Table 3, we find greater total delegation to politically responsive agencies (on the left-hand side of the figures) during unified government, and relatively greater delegation to politically independent agencies during times of divided government, though estimated delegations are less for both regime types as agencies become more independent.

These findings are consistent with the observations from Vannoni *et al.* (2021) (for the states), which shows that Congress is more willing to delegate more to politicized agencies during unified government than during divided government. This also fits with findings by Epstein & O’Halloran (1999) and Volden (2002) that Congress gives independent agencies more discretion during divided government.

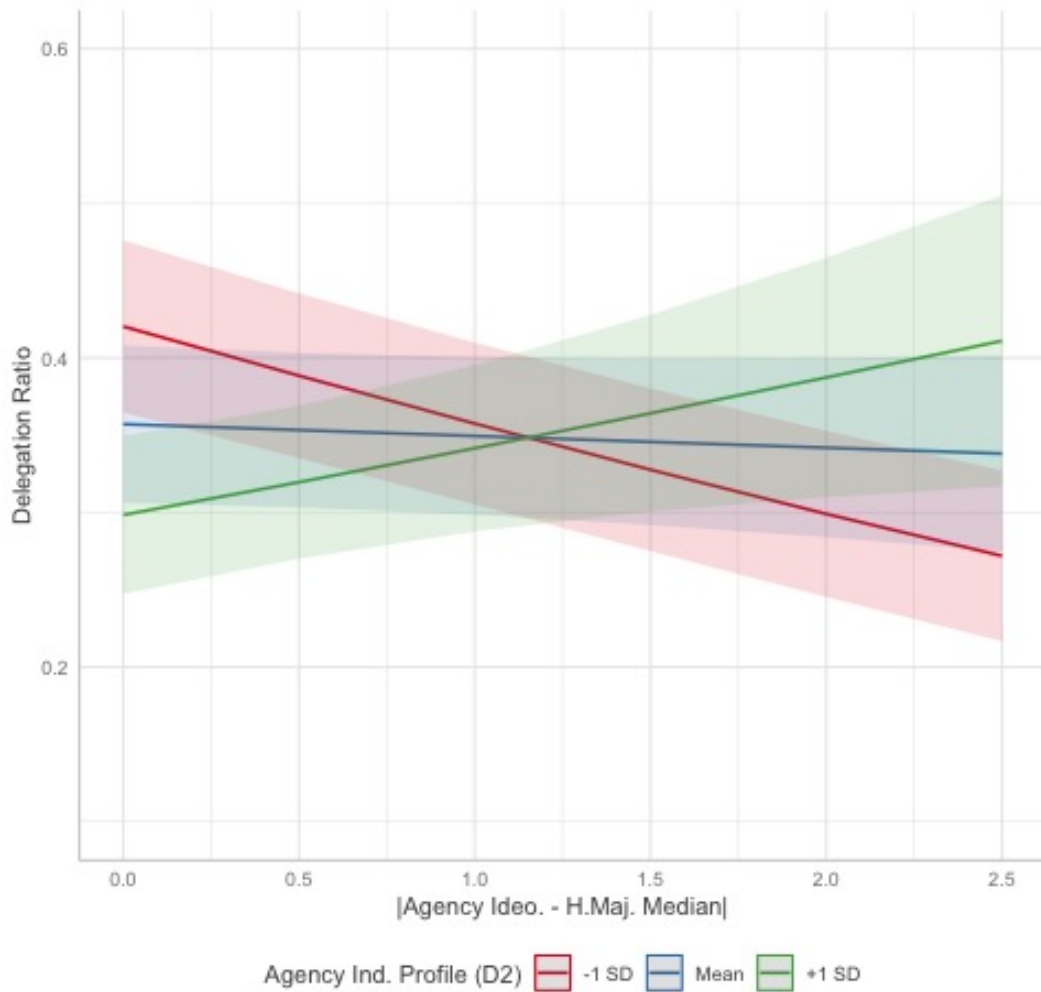
The coefficient on the interaction term between ideological distance and agency independence is positive in all models presented in Table 3, and statistically significant in the model fit on bills that passed the House. We plot the marginal mean effects from this model in Figure 4 to further explore this result. Interestingly, while the classic ally principle expectation—that delegation becomes less attractive to the principal as the agent becomes more ideologically distant—is borne out for highly-politicized agencies, a different relationship exists for more independent agencies. Specifically, for agencies one standard deviation above the observed mean of Selin’s second dimension independence score, we see a pattern indicative of a reverse ally principle, in which Congress delegates marginally more to agencies as they become increasingly ideologically distant. This particular finding is consistent

Figure 3: Conditional Effect of Agency Independence from Political Review on Delegation



with work on delegation and credible commitments (e.g., Knott & Miller 2008; Gersen & Vermeule 2012; Miller & Whitford 2016), which argues that principals occasionally prefer to delegate to ideologically distant and insulated agents in order to make their commitment to some policy more credible. Overall, these results demonstrate how specific ideological arrangements and structural conditions affect delegating behavior.

Figure 4: Agency Independence, Agency Ideology, and Delegation



Discussion

In this paper, we have demonstrated how an active learning convolutional neural network for classifying text can be used to study a complex problem in political science: agency delegation from Congress. We hope that the proposed methods are straightforward and usable for other applications and that gains in classification accuracy, while reducing the need for extra documents to be hand-labeled, helps researchers tackle challenging classification problems. This information is part of an ongoing endeavor to learn how Congress uses statutory language to enact its agenda and how the modern legislative process provides

oversight and guidance to implement policies. These classifications will help provide the means through which we can test our theories of delegation and congressional oversight on a larger scale and provide the nexus for increased research into the implications of statutory language.

We then use our measure of delegation to gain new leverage on an important theoretical proposition and empirical finding throughout the delegation literature—the ally principle. Because our text-based classification model allows us to measure delegation in a much larger and more inclusive set of legislation than has been used in previous work, we are able to move beyond a focus on landmark legislation and speak to the generalizability of previous findings (Epstein & O’Halloran 1999, pg. 89). Additionally, our use of agency-specific independence and ideology measures (Selin 2015; Clinton *et al.* 2012), along with our focus on a time period that saw a transition from divided to unified government, allows us to better isolate the effect of presidential administration on congressional delegation decisions. We find that the majority party in Congress typically prefers to delegate more to ideologically-proximate and politically-responsive agencies, regardless of whether the president is a co-partisan or not. Our findings suggest that congressional delegation decisions are driven by a legislative view of the executive branch as a pluralistic internally-complex institution, the various parts of which are not perfectly under presidential control.

We expect that deep and active learning models of text classification can be used in various ways to further our knowledge of delegation. While we lay out in this paper some concerns about the measurement of discretion, it would be possible to apply our general method to the systematic identification of the categories of constraints laid out in Epstein & O’Halloran (1999), which would then allow for some measurement of discretion. Recent work by Lowande & Shipan (2021) has made important strides towards the identification of discretion as it is wielded by the president in certain policy areas, but this is a separable task from identifying the *sources* of this discretion. Our methodology could be helpful in identifying and measuring discretion as it exists in legislative texts, thereby pinning down

the quantity of interest in many studies of delegation. Additionally, because our approach can be used to identify delegation in all versions of all bills, it can gain empirical leverage on theoretical propositions about delegation throughout the legislative process. For example, expectations derived from Bawn (1997) about how the level of delegation may change as bills are considered in committee and on the floor could be tested with our measure.

Furthermore, there are multiple potential applications of our methodology beyond delegation. In particular, we believe that problems involving the classification of complex texts, particularly those requiring expertise for labeling, can benefit from schemes similar to our method. Any problem that involves classifying complex texts and requires context will benefit from a deep learning solution. Moreover, any classification problem requiring particular knowledge (or is improved by learned expertise) would benefit from active learning. We envision this as the first of many applications of this model to the coding of legislative texts for a number of important features with potential implications for interbranch relations, policymaking, and legislative bargaining.

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